H2AN: Hierarchical Homogeneity-Attention Network for Hyperspectral Image Classification

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***Abstract*— Recently, a self-attention network (SAN) is developed as an effective strategy to extract features from attention areas for image classification. However, for hyperspectral image (HSI), the lack of position supervision of object regions and inefficient similarity computation lead to unsatisfactory classification performance on mixed pixels. To alleviate the above two problems for HSI image classification, we propose a novel hierarchical homogeneity-attention network (H2AN) in this article. First, we design a homogeneity-attention block (HAB) to depict the feature correlation with the homogeneous mask. Using the supervision of homogeneity mask, we can calculate the attention guided by the predefined number of homogeneity embeddings, which can reduce the heavy computation instead of the global search in self-attention block (SAB) of SAN. Second, we propose a hierarchical convolutional neural network (HCNN) inserting the HAB into different levels of network cells for highly efficient feature extraction of target regions, named H2AN. Because of the transferring of homogeneity property from shallow layer to deep layer, our H2AN outperforms the state-of-the-art methods in qualitative and quantitative experiments on three typical datasets.**

***Index Terms*— Convolutional neural network (CNN), homogeneity, hyperspectral image (HSI) classification, self-attention network (SAN).**

I. Introduction

HYPERSPECTRAL image (HSI) classification is a significant research direction. Due to the rich spatial and spectral information, HSI classification is widely employed in many remote fields, such as land-cover mapping, land change monitoring, and ecological science [1]–[6]. With the development of the machine learning technique, the existing HSI classification methods can fall into two types: traditional machine learning and deep learning methods.

Traditional machine learning methods for HSI classification usually focus on classifier design with different feature extractions [7]. Based on the original image features, Blanzieri and Melgani [8] proposed a k-nearest neighbor (k-NN) remote sensing classification method using the Euclidean distance to measure the similarity between a testing sample and the available training samples. Licciardi *et al.* [9] combined nonlinear principal component analysis (PCA) feature extractor and supported vector machine (SVM) classifiers [10] to improve the classification accuracy. Jia *et al.* [11] proposed a collaborative representation based on the 3-D Gabor feature to obtain higher performance in terms of both the classifier complexity and generalization ability on very small training sets. Usually, these machine learning methods only focused on feature extraction strategies within some classical classifiers. On the other hand, to consider the interaction between feature extraction and classifier design, sparse representation (SR) was employed to learn the sparse coefficients as features and comprehensively measured the construction error for the final HSI classification. With both spatial constraint and sparse constraint, Chen *et al.* [12] first proposed a joint sparse model (JSM) to extract sparse features of the hyperspectral pixels. Based on JSM, Gao *et al.* [13] introduced the discontinuity preserving relaxation to weigh the contribution from each neighborhood pixel. With these sparse coefficients, the representation error was used as the final classification criterion instead of traditional explicit classifiers.

Moreover, with the rapid development of the deep learning technique [14], recent HSI classification approaches emerged to extract high-level features and train a good classifier. Via end-to-end learning, they could obtain better classification performance than the abovementioned traditional methods.

Initially, each HSI pixel was viewed as independent spectral signatures to input into deep networks, such as deep belief network (DBN) [15], stacked autoencoder (SAE) [16], and recurrent neural network (RNN) [17]. Only feeding simple HSI pixel signatures, no matter the supervised DBN and RNN or unsupervised SAE models cannot obtain enough accurate results. In order to extract more convincing features, some researchers developed deep learning-based models via integrating spatial information into spectral signatures. By taking the lead, Chen *et al.* [18] presented a simple convolutional neural network (CNN) framework by combining the idea of AlexNet [19] and 3-D convolution kernels to capture spectral–spatial features for HSI classification. To extract deeper features, Lee and Kwon [20] proposed a contextual deep CNN in virtue of the inception module, such as

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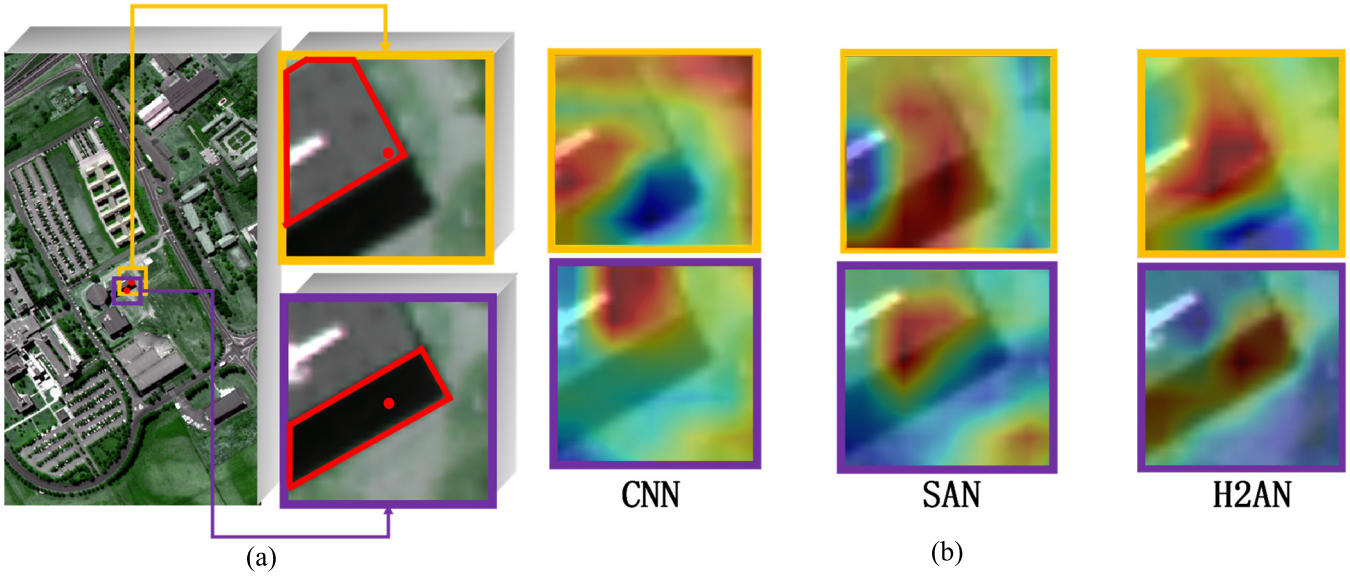


Fig. 1. Mixed HSI cube examples of the PU dataset. (a) Two input mixed image cubes on the original HSI dataset. (b) CAMs of CNN [18], SAN [26], and our proposed H2AN.

GoogLeNet [21] and the skip architectures from ResNet [22]. Based on this, their network training could be still remaining stable even if its depth reaches nine layers deeper than many state-of-the-art CNNs for HSI classification in that era [20]. Afterward, the idea of ResNet motivated the publications of a range of deep learning frameworks for HSI classification. For example, Zhong *et al.* [23] presented a spectral–spatial network, which is made up of unified sequential residual convolutional blocks. Paoletti *et al.* [24] upgraded the residual blocks to pyramidal bottleneck residual blocks in order to enrich the diversity of high-level features in deeper layers for better performance. Moreover, Zhang *et al.* [25] designed a multiscale dense network to integrate scale features in different depths by means of dense connection, which is regarded as a better version of skip connection. In addition, graph convolutional networks [26] were integrated into CNNs to enhance the ability in modeling relations between the samples.

As we have known, the abovementioned models could obtain good performance under the assumption that the whole input cube is labeled by the center pixel in the deep learning HSI classification framework. Recently, Sun *et al.* [27] pointed out that the mixed HSI cube disturbs the discriminative feature extraction due to the class label inconsistency between the pixels at the boundary and center positions. Taking a toy example, as shown in Fig. 1, two mixed regions of the Pavia University (PU) dataset are marked with orange and purple boxes, respectively, and two target regions with the same labels to the red center pixels are highlighted by the red borderline in the thumbnails of Fig. 1(a). To interpret the importance distribution, we visualize their corresponding class activation mappings (CAMs) in Fig. 1(b), under different models including the previously mentioned CNN [18], self-attention network (SAN) [27], and our proposed hierarchical homogeneity-attention network (H2AN). Compared with the original CNN from the CAMs in Fig. 1(b), it can be obviously observed that SAN can help CNN focus on the part of target regions since SAN employs a self-attention block (SAB) to leverage features from all locations in an image cube. This SAB module improves the ability of CNN to capture global information, which is beneficial to the target region's feature extraction. However, the feature extraction of SAB is still inefficient in the following two aspects: 1) limited ability to extract features from the target region; SAB is hard to grab the exact location of the target region due to the lack of extra target's prior information and 2) heavy computation cost. SAB computes the similarities of all locations between each other, requiring a matrix multiplication of computational complexity $\mathcal{O}(ch^2w^2)$, given an input feature map with size $c \times h \times w$. These problems can be solved in our proposed H2AN, as shown in Fig. 1(b). To grab more comprehensive target regions, we design a novel and efficient attention module named homogeneity-attention block (HAB). Specifically, the homogeneous mask is introduced as extra input to assist the network is focusing on the target's region, which can enhance the target's feature extraction ability. Then, the homogeneity embeddings are utilized to guide the selection of homogeneity representative pixels as the similarity comparison objects. This improvement can reduce the attention computational complexity from $\mathcal{O}(ch^2w^2)$ to $\mathcal{O}(chwp)$, where p is the predefined number of homogeneity representative embeddings. For highly efficient feature extraction, we fuse the HAB module into different levels of HSI classification network cells, which integrates our H2AN. It is noteworthy that hierarchical architecture can transfer homogeneity supervision from a shallow portion to the following parts, which can further improve the feature extraction ability.

The main contributions of this article can be summarized as follows.

1) We design a novel attention module HAB to improve efficiency in extracting target features, by adding

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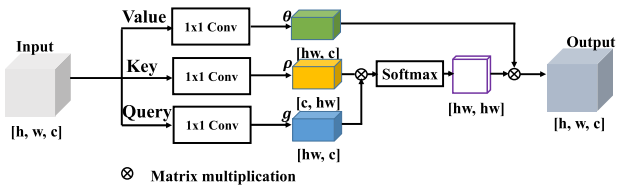


Fig. 2. Architecture of the SAB.

homogeneous masks as extra supervision and simplifying the similarity matrix computation.

2) We propose an H2AN with HAB to transfer the target homogeneity property from shallow layer to deep layer, which not only drives each layer to extract the target's feature but also improves the performance of the network in HSI classification.
3) We carry out extensive experiments on three typical hyperspectral datasets: Indian Pines (IP), PU, and Pavia Center. Besides the conventional accuracy and classification map comparisons, we visualize the CAMs to furthermore prove that H2AN is more efficient to focus on the target region. Via the appropriate and hierarchical insertion of HAB, H2AN can pay attention to extract more discriminative features compared to the CNN model and the SAB model. Quantitative and qualitative results demonstrate that our proposed network outperforms several state-of-the-art methods.

The remainder of this article is structured as follows. Our H2AN is introduced in detail in Section II. The effectiveness of the proposed method is demonstrated in Section III on three real HSIs. Finally, Section IV concludes this article.

II. Our Method

Our H2AN aims at inserting corresponding homogeneity attention HAB into different network cell levels of a hierarchical CNN framework. In particular, the HAB is designed to aggregate target features, which is first described in detail as follows.

A. Homogeneity Attention Block

To rich the interpretation of feature extraction, we concentrate on the development of attention mechanisms, specifically considering the corresponding homogeneous mask to guide the attention calculation. In fact, it is inspired by the work of SAB [28]. The original SAB leverages features from all pixels equally in one image cube, as shown in Fig. 2.

Denoted the spatial size and the channel number as $h \times w$ and c, the input feature map of the block is represented as $\boldsymbol{x} \in \mathbb{R}^{h\times w\times c}$. First, $\boldsymbol{x}$ is converted to $\theta(\boldsymbol{x})\epsilon\mathbb{R}^{h\times w\times c}$, $\rho(\boldsymbol{x})\epsilon\mathbb{R}^{h\times w\times c}$, and $g(\boldsymbol{x})\epsilon\mathbb{R}^{h\times w\times c}$ by implementing convolution layers with a kernel size of 1×1, where c represents the number of convolution kernels. Specifically, the transformations can be written as

$$\theta(\boldsymbol{x}) = \boldsymbol{x}\boldsymbol{W}_{\boldsymbol{\theta}}, \quad \rho(\boldsymbol{x}) = \boldsymbol{x}\boldsymbol{W}_{\boldsymbol{\rho}}, \quad g(\boldsymbol{x}) = \boldsymbol{x}\boldsymbol{W}_{\boldsymbol{g}} \tag{1}$$

where $\boldsymbol{W}_{\boldsymbol{\theta}}$, $\boldsymbol{W}_{\boldsymbol{\rho}}$, $\boldsymbol{W}_{\boldsymbol{g}} \in \mathbb{R}^{1\times 1\times c\times c}$ are learnable weight matrices. Second, $\theta(\boldsymbol{x})$ and $g(\boldsymbol{x})$ are reshaped into $R^{hw\times c}$, while $\rho(\boldsymbol{x})$ is reshaped into $\mathbb{R}^{c\times hw}$, where hw is the product of the height and width of feature map $\theta(\boldsymbol{x})$. Third, the relation matrix $\mathbf{R} \in \mathbb{R}^{hw\times hw}$ of all pixels in the input $\boldsymbol{x}$ is defined according to the following equation:

$$\mathbf{R} = \theta(\boldsymbol{x})\rho(\boldsymbol{x}). \tag{2}$$

To normalize $\mathbf{R}$ to a range of 0–1.0, the softmax function is implemented on each row of $\mathbf{R}$, which is denoted as

$$\widehat{\boldsymbol{R}}(i, j) = \frac{e^{R(i,j)}}{\sum_{j=1}^{hw} e^{R(i,j)}}. \tag{3}$$

Next, the attention features $\boldsymbol{A} \in \mathbb{R}^{hw\times c}$ is the relation weighted sum of the features at all positions

$$\boldsymbol{A} = \widehat{\boldsymbol{R}}g(\boldsymbol{x}). \tag{4}$$

The SAB is designed to capture the attention features in theory. However, it is without extra position supervision of target regions, so it is hard to focus on the target region in feature extraction. In addition, its global similarity requires a heavy computation load. Thus, its performance is still not enough in the HSI classification task. To ease the above problems, we propose a new structure of attention module HAB, as shown in Fig. 3. Extending from the SAB, HAB is added homogeneous mask and homogeneity embeddings selection. Considering that the feature representation becomes more abstract when the layer getting deeper, there are original image pixels and learned feature maps as the source of the homogeneous mask. Therefore, homogeneous masks can be generated in two ways from the original image pixel and learned feature map, which constructs two branches for HAB.

For the original image pixels, we deal with the superpixel map, in which p positions are randomly selected from the center pixel's homogeneous region as 1, and the rest positions are randomly selected from the center pixel's homogeneous region as 0. It is noteworthy that the superpixel map is obtained by a space-clustering algorithm, namely, simple linear iterative clustering (SLIC) [29] in this article. It can be supposed that the pixel marked by the red point is the center pixel, and the homogeneous region consists of pixels with the same color as the center pixel. Considering that the superpixel segmentation is not enough robust to various land covers, we set a smaller p in the first level of HAB, which can pay some attention to homogeneous regions (superpixel map) without bringing heterogeneous interference.

For the learned feature map, we first derive an activation map from the mean value of each row of the previous layer's output tensor in the given channel dimension. Then, in the activation map, we select the top p activation positions in the activation map as 1, and the rest positions are set 0. It can be assumed that the positions with high activation values in the feature map are regarded as the clustering centers of the abstract features. Therefore, it is suitable to set the value of p slightly higher in the second and third levels of HAB to pay much attention to the homogeneous feature.

These two branches can be selected by a switch according to the input types. Therefore, the input data are through different branches at different layer levels. Based on the homogeneity map, p points are selected from the original

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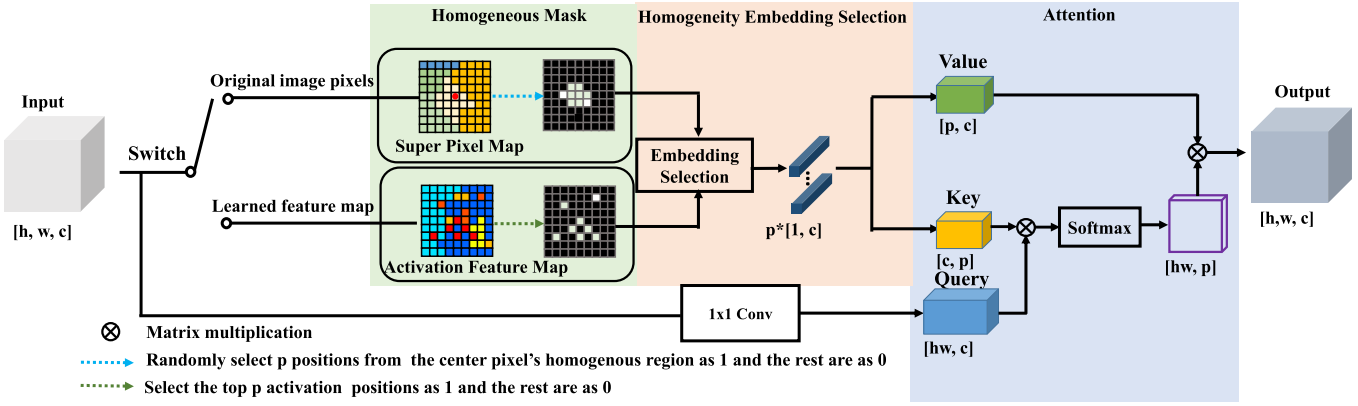


Fig. 3. Architecture of the proposed HAB.

input to construct the homogeneity embeddings, which are reshaped into $\mathbf{E} \in \mathbb{R}^{p\times c}$, in the attention process. To decrease the parameter size, we only transform the input feature map $\boldsymbol{x}$ into $\theta(\boldsymbol{x})$ with one convolution layer via c convolutional kernels of size 1×1. Then, it is noteworthy that only the relation matrix $\boldsymbol{R}' \in \mathbb{R}^{hw\times p}$ is required to consider computing between all positions with homogenous region embeddings, which is defined as

$$\boldsymbol{R}' = \theta(\boldsymbol{x})\boldsymbol{E}^{\mathrm{T}}. \tag{5}$$

Simultaneously, the softmax function is utilized to normalize each row of $\boldsymbol{R}'$, which is expressed as

$$\widehat{\boldsymbol{R}}'(i,j) = \frac{e^{R(i,j)}}{\sum_{j=1}^{p} e^{R(i,j)}}. \tag{6}$$

In this way, novel attention features $\boldsymbol{A}'\epsilon\mathbb{R}^{h\times w\times c}$ are formulated as

$$\boldsymbol{A}' = \widehat{\boldsymbol{R}}'\boldsymbol{E}. \tag{7}$$

To further demonstrate the efficiency of the proposed HAB, we briefly give a computational efficiency comparison with SAB. As illustrated above, there are three learnable transformations, namely, $\boldsymbol{W}_{\theta}$, $\boldsymbol{W}_{\rho}$, and $\boldsymbol{W}_{g}$ in the SAB. These convolutional operations are designed to project the original input to another embedding space for more generalization. The total computation cost for them can, thus, be written as

$$\mathrm{Cost}_{\mathrm{SAB,conv}} = 3 \times c^2 \times hw \times c = 3c^3hw. \tag{8}$$

The following matrix multiplication of the embeddings from conv__θ and conv__ρ is expressed as follows:

$$\mathrm{Cost}_{\mathrm{SAB,matmul}} = c \times (hw)^2 = c(hw)^2. \tag{9}$$

The subsequent matrix multiplication between the softmax output of the above matrix multiplication and the embeddings from conv__g take the same computation cost. Thus, the total computation of the SAB could be presented as

$$\begin{aligned}\mathrm{Cost}_{\mathrm{SAB}} &= \mathrm{Cost}_{\mathrm{SAB,conv}} + 2 \times \mathrm{Cost}_{\mathrm{SAB,matmul}} \\ &= 3c^3hw + 2c(hw)^2.\end{aligned} \tag{10}$$

Simultaneously, the total computation cost of our proposed HAB can be easily calculated as

$$\begin{aligned}\mathrm{Cost}_{\mathrm{HAB}} &= \mathrm{Cost}_{\mathrm{HAB,conv}} + 2 \times \mathrm{Cost}_{\mathrm{HAB,matmul}} \\ &= c^3hw + 2pchw.\end{aligned} \tag{11}$$

Although the SLIC algorithm brings about some computation cost, it does virtually nothing to increase the computation cost of the HAB. From the analysis of computation cost, it could be proven that our proposed HAB is more efficient than the SAB in terms of lower computation cost.

B. Hierarchical Homogeneity-Attention Network

Surrounded around the theme of how to extract more discriminative features, a series of models are proposed to extract multiscale features (the contextual CNN [20]), deformable features (the deformable CNN [30]), hierarchical features (the deep feature fusion network [31]), and attention features (the hard attention generative adversarial networks (GAN) [32] and the multibranch CNN [33]). Due to the effective classification performance, we settle down to the development of the baseline CNN model. In addition, to make the feature extraction of target regions more efficient for HSI classification, we propose a hierarchical CNN (HCNN) inserting the above proposed HAB into different levels of network cells, named H2AN.

1) Overall Description: As shown in Fig. 4, our proposed H2AN includes the input cell, three-level features extraction cells, and the final classification cell. Besides, we give drawing statements at the bottom of Fig. 4. Specifically, the bottle-green vertical cube stands for the 3×3 convolutional layers following by a Relu activation layer. The blue–black vertical cube and rectangle represent the 2-D batch normalization layer and 1-D batch normalization layer individually. Deep orange vertical cube and yellow one indicate the maxpooling layer and the global average pooling layer, respectively. The orange rectangle and gray one denote one fully connected layer and the softmax layer severally. Next, we describe these components of H2AN in detail, which are comprised of these colored cubes or rectangles.

2) Input Cell: Following the abovementioned CNN models for HSI classification, the original HSI data is cropped into

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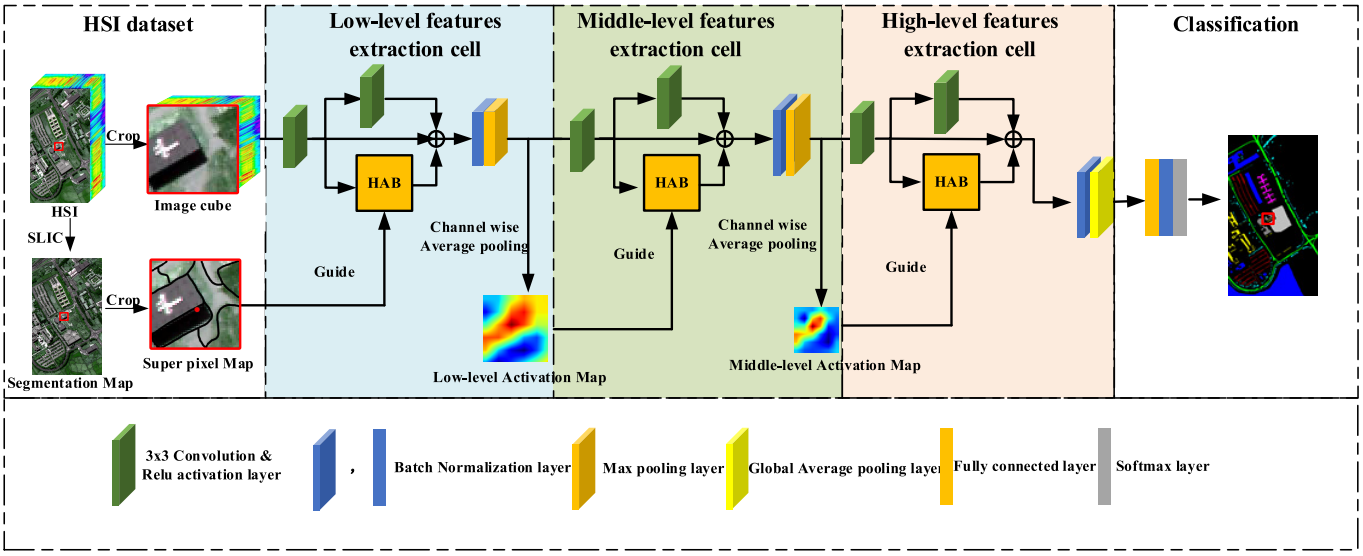


Fig. 4. Architecture of the proposed H2AN.

patches labeled according to the corresponding center pixels just like the red point in an image cube shown in Fig. 4. Then, to initialize the supervision of homogeneous mask, we obtain a segmentation map $\boldsymbol{H^1} \epsilon \mathbb{R}^{h \times w}$ by implementing SLIC [29] on the original HSI.

3) Features Extraction Cells: To transfer homogeneity property from the shallow layer to the deep layer, three-level feature extraction cells form the main components of H2AN. They are low-, middle-, and high-level feature extraction cells. Their main function is to help the network focus on the discriminative feature extraction of target regions more effectively and efficiently.

Inspired by the idea of inception [21], their same structure includes three different branches to capture diverse features for better classification performance. The first branch includes two successive convolution layers with 3×3 kernels following the Relu function to extract large-scale convolutional features, which is equal to 5×5 convolutional kernel's receptive fields [34]. The second branch contains one convolutional layer with 3×3 kernels following the Relu function to extract the small-scale features, which is conducive to the convergence of H2AN. It is noteworthy that the first branch and the second branch share one convolutional layer to lighten the total model parameters of H2AN. The third branch contains one 3×3 convolution layer with the Relu function and the proposed HAB to obtain the discriminative target attention features, which forces the corresponding cell to focus on the important target region. In addition, the add operation on the outputs of three branches enables the cell to obtain diverse features that fuse attention features and multiscale convolution features.

Note that the guidance of HAB in these cells is different. The low-level's HAB adopts the superpixel map as the first level of homogeneity guidance, which remains the original spatial constraint of the homogeneity region. The middle- and high-level HABs receive their previous cells' activation maps as the second and third levels of homogeneity guidance. The activation map is derived via the channelwise averaging pooling to emphasize the spatial salient region as the following cell's homogeneous mask. Specifically, given a feature map $\boldsymbol{x} \in \mathbb{R}^{h \times w \times c}$, the value H_{ij}^2 in the (i, j) activation map $\boldsymbol{H^2} \epsilon \mathbb{R}^{h \times w}$ is denoted as

$$H_{ij}^2 = \frac{\sum_{k=1}^{k=c} x_{ijk}}{c}. \tag{12}$$

In order to speed the converge of the network, batch normalization is employed in each cell. As important parts for further extracting high-level features, two max-pooling operations connect the three cells end to end.

4) Classification Cell: When the highest-level features are obtained, the global average pooling operation, one fully connected layer with batch normalization, and the softmax function are utilized for HSI classification. The softmax function outputs the prediction label vector $\hat{y}_i = [\hat{y}_i^1, \hat{y}_i^2, \ldots, \hat{y}_i^N]$. The final objective function is formulated as

$$L = -\frac{1}{b} \sum_{i=1}^{b} \sum_{j=1}^{N} \mathbf{1}(y_i = j) \log \hat{y}_i^j \tag{13}$$

where y_i is the true label of the ith dataset, b represents the batch size in one training iteration, $\mathbf{1}(y_i = j)$ is the indicator function (if $y_i = j$, $\mathbf{1}(y_i = j)$ is 1; else, $\mathbf{1}(y_i = j)$ is 0), and N denotes the total number of land-cover categories.

III. Experimental Results

A. Experimental Data and Quantitative Evaluation Index

In order to demonstrate the performance of the proposed H2AN, we test some experiments on three typical datasets, including, IP, PU, and Pavia Center. The IP image was obtained by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS), which is collected over the Northwestern Indiana agricultural test site in June 1992. It has 145×145 pixels with a spatial resolution of about 20 m per pixel. The number of spectral bands is 224, ranging from 0.4 to 2.5 μm. In our

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TABLE I
NUMBER OF RANDOMLY SELECTED PIXELS FOR THE IP, PU, AND PCS

Class	Indian Pines (5%)			Pavia University(1%)			Pavia Center(2%)		
	Name	Train	Test	Name	Train	Test	Name	Train	Test
1	Alfalfa	3	51	Asphalt	68	6784	Water	1306	63972
2	Corn-notill	72	1362	Meadows	187	18499	Trees	130	6377
3	Corn-mintill	41	783	Gravel	22	2185	Meadows	58	2833
4	Corn	12	222	Trees	34	3402	Blocking Bricks	43	2109
5	Grass-pasture	25	472	Metal sheets	14	1364	Bare Soil	131	6418
6	Grass-trees	37	710	Bare Soil	51	5053	Asphalt	151	7374
7	Grass-pasture-mowed	1	25	Bitumen	14	1342	Bitumen	151	7406
8	Hay-windrowed	24	465	Blocking Bricks	39	3839	Tiles	62	3060
9	Oats	1	19	Shadows	10	1016	Shadows	43	2122
10	Soybean-notill	48	510	—	—	—	—	—	—
11	Soybean-mintill	123	2345	—	—	—	—	—	—
12	Soybean-clean	31	583	—	—	—	—	—	—
13	Wheat	11	201	—	—	—	—	—	—
14	Woods	65	1229	—	—	—	—	—	—
15	Buildings-Grass	19	361	—	—	—	—	—	—
16	Stone-Steel-Towers	5	90	—	—	—	—	—	—
	Total	518	9838	Total	439	43484	Total	2070	101406

experiment, we only use the remaining 200 spectral bands after removing the 24 water absorption bands. The overall image covers 16 classes of land features, which are explained in Table I. Its three-band pseudocolor composite image, ground truth, and color code are shown in Fig. 5. The PU was captured by the Reflective Optics System Imaging Spectrometer (ROSIS), which has 103 spectral wavelengths ranging from 0.38 to 0.86 μm. Its spatial size is 610×340 with 1.3-m spatial resolution covering nine classes of ground objects whose names are shown in Table I. The false-color composite, ground truth, and color code are shown in Fig. 6. The Pavia Center image was also captured by the ROSIS sensor over an urban area surrounding the Center of Pavia, Italy. The Pavia Center dataset (PC) consists of 1096×492 pixels, each including 103 bands with the 12 noisy bands removed. Nine classes of land covers are considered, which is described in Table I. Fig. 7 illustrates the false-color composite image and the corresponding ground reference map.

In order to demonstrate the classification performance, the overall accuracy (OA) and the Kappa coefficient are selected as the classification indexes. The OA can be acquired by computing the percentage of correctly classified test pixels. The Kappa coefficient shows the percentage of classified pixels corrected by the number of agreements that would be expected purely by chance. For these two classification indexes, the larger value means the better performance.

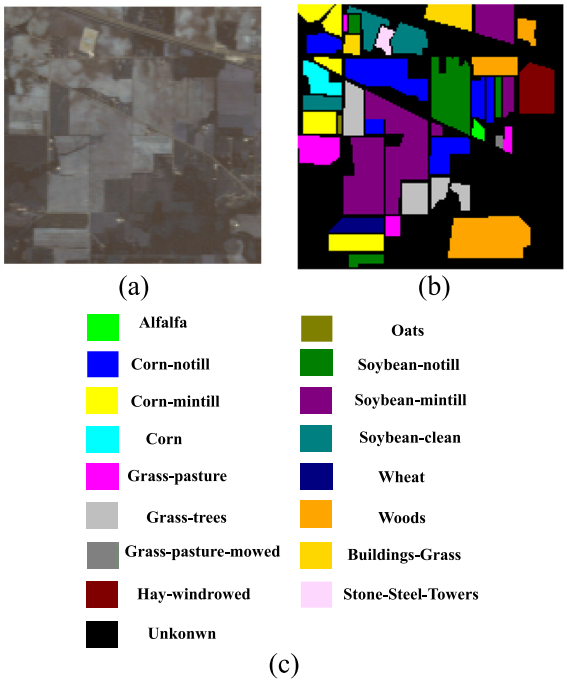


Fig. 5. IP dataset. (a) Three-band false color composite. (b) Ground truth. (c) Color code.

B. Implementation Details and Framework Setting

To find the most appropriate network architecture, four different cases are explored in three datasets. The three cases measure the feature levels where HABs are embedded, the connection types between the HAB, the convolution layers and the number of p in each level of HAB, and the spatial size of input data.

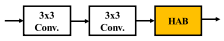
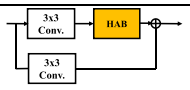
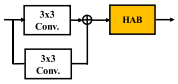
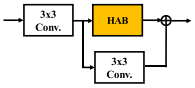
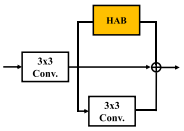
First, the case considering HABs at different levels of features is compared by conducting some experiments on three datasets. The performance comparison is reported in Table II. As shown in Table II, "HAB-L," "HAB-M," and "HAB-H" represent that only one HAB is inserted into the low-level cell, the middle-level cell, or the high-level cell, respectively. "HAB-L + M," "HAB-L + H," and "HAB-M + H" denote that two HABs are embedded into the low-level cell and the middle-level cell, the low-level cell and the high-level cell, and the middle-level cell and the high-level cell, respectively. Furthermore, "HAB-L + M + H" represents that the three feature levels are all embedded with corresponding HAB. Meanwhile, "HAB-WO" indicates these cells without HAB, which is taken as a baseline. As shown in Table II, one HAB insertion,

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TABLE II
OAs (%) OF THE HAB WITH DIFFERENT LEVELS OF FEATURES

	Indian Pines	Pavia University	Pavia Center
HAB-WO	93.5±1.4	94.1±0.3	98.5±0.5
HAB-L	96.4±0.6	96.0±0.7	99.2±0.4
HAB-M	96.2±0.1	95.9±0.4	99.1±0.1
HAB-H	96.1±0.2	95.5±0.3	99.1±0.1
HAB-L+M	96.7±0.1	96.7±0.2	99.3±0.5
HAB-L+H	96.7±0.0	96.5±0.1	99.3±0.1
HAB-M+H	96.6±0.2	96.5±0.1	99.3±0.1
HAB-L+M+H	**97.2±0.5**	**97.4±0.6**	**99.7±0.1**

TABLE III
OAs (%) WITH DIFFERENT TYPES OF CONNECTIONS BETWEEN THE HAB AND THE CONVOLUTION LAYERS

Type	Block	Indian Pines	Pavia University	Pavia Center
1		96.6±0.1	96.9±0.2	97.7±0.3
2		96.4±0.3	96.6±0.1	97.4±0.2
3		97.0±0.2	97.1±0.0	98.9±0.1
4		96.9±0.2	97.0±0.1	98.8±0.1
5		97.1±0.5	97.2±0.2	99.3±0.3
6		**97.2±0.5**	**97.4±0.6**	**99.7±0.1**

including HAB-L/M/H, can improve the performance of the basic network HAB-WO. Moreover, more HABs no matter inserting in two levels or three levels bring more accuracy improvement, and the HAB-L + M + H inserting at all three levels obtains the best OA results on three datasets because a variety of feature extraction is taken into account.

Second, to explore the impact of the connection types between the HAB and the convolution layers, we design six connection types shown in Table III. This is done because it is improper to insert the HAB into the first layer in the cell, considering that the bands of the input cube are large, which increases the calculated amount according to (11). Their corresponding performance on three datasets is illustrated in Table III. It can be illustrated that the sixth type achieves the best classification performance on three datasets. Thus, we adopt the sixth type of connection as the basic connection of each cell, as can be seen in Fig. 3.

TABLE IV
p OF THE HAB IN DIFFERENT LEVELS OF CELLS

	Indian Pines	Pavia University	Pavia Center
Low-Level p	27	380	427
Middle-Level p	35	100	110
High-Level p	30	35	35

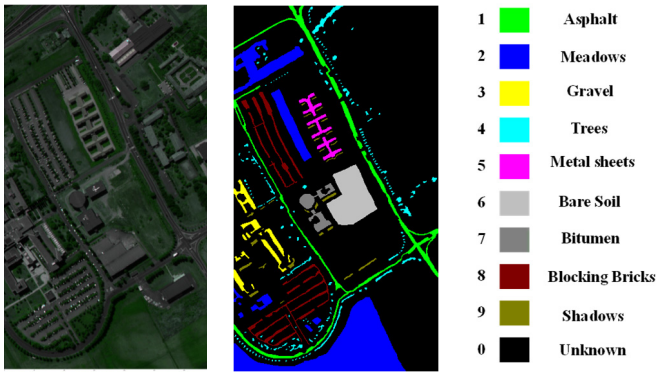


Fig. 6. PU dataset. (a) Three-band false color composite. (b) Ground truth. (c) Color code.

Third, the value of p represents the number of homogeneity embedding features that have a great effect on the performance of the proposed H2AN. In the first-level cell, we counted the pixel number of center pixel's homogeneity region in all superpixel maps. To meet the requirement of parallel operation, we set p in the first level of HAB as the average of the abovementioned statistical figures. The spatial sizes of the low-level feature map and the middle-level feature map are 13×13 and 6×6. Via trial-and-error analysis, p in middle-level HAB is tuned from 5 to 165, and p in high-level HAB is tuned from 5 to 35. Finally, p in each-level HAB for three datasets is shown in Table IV.

Finally, the spatial size of cropped HSI patches, indeed, has a great influence on classification results. A larger spatial size means that more spatial information could be utilized, while more mixed pixels are involved in feature extraction. Several experiments are conducted to determine the most appropriate spatial size on three datasets. The spatial size is set from 21×21 to 33×33. The classification results with different sizes of input data are presented in Fig. 8. It can be seen that the classification accuracy on three datasets increases at first and then decreases with the increase in the spatial size. In addition, our proposed model achieves the best classification accuracy on three datasets when the size is 27×27. Thus, we select the 27×27 as the fixed spatial size of the input cube in our H2AN. To better understand the network architecture of H2AN on three datasets, the detailed structures and hyperparameters are summarized in Table V.

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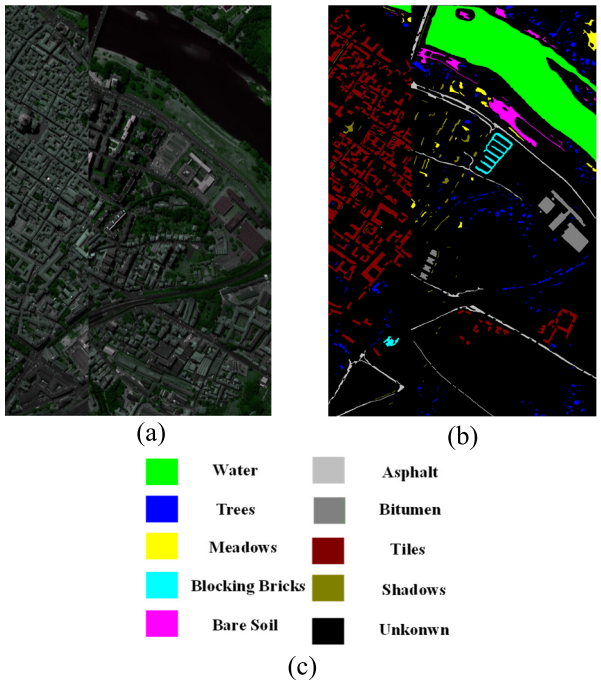


Fig. 7. PC. (a) Three-band false color composite. (b) Ground truth. (c) Color code.

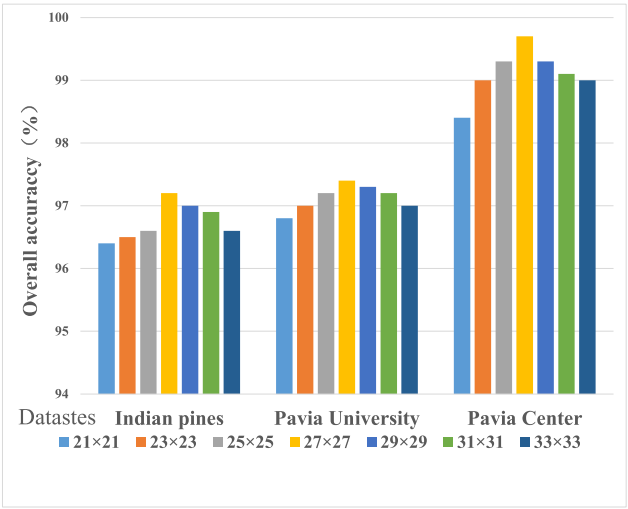


Fig. 8. Influence of spatial size on the (Left) IP, (Middle) PU, and (Right) Pavia Center. Best viewed in color.

C. Comparing With Other Methods

To demonstrate the performance of the proposed H2AN, the qualitative and quantitative classification results of H2AN are compared with two traditional CNN-based HSI classification methods (Conv-2D and Conv-3D) [18], two recently representative Capsule-based methods (Conv-Caps [35] and CapsNet [36]), and two representative attention-based methods (SAN [27] and joint spatial–spectral attention network (JSSAN) [37]) published in recent one year for comparison. In addition, the baseline models, HCNN and HCNN with self attention (HCNN-SA), with SAB are also evaluated in this experiment. For a fair comparison, the proposed method and compared methods adopt the same experiment setups

TABLE V
DETAILED STRUCTURES AND HYPER PARAMETERS OF H2AN ON THREE DATASETS

	Layer	Kernel number	Indian Pines	Pavia University	Pavia Center
Input size	-	-	27×27	27×27	27×27
L-L cell	2D-Conv	16	✓	✓	✓
	2D-Conv	16	✓	✓	✓
	HAB	16	✓	✓	✓
M-L cell	2D-Conv	32	✓	✓	✓
	2D-Conv	32	✓	✓	✓
	HAB	32	✓	✓	✓
H-L cell	2D-Conv	64	✓	✓	✓
	2D-Conv	64	✓	✓	✓
	HAB	64	✓	✓	✓
GAP	-	-	✓	✓	✓
FC	-	n	✓	✓	✓
Cell types	-	-	6	6	6
p	-	-	27/35/30	380/100/35	427/110/35

1. n denotes the total number of land-cover classes.
2. Cell types represent the connection types between HAB and convolutional layers which is shown in Table III.
3. FC denotes the fully connected layer.
4. GAP denotes the Global Average Pooling layer.
5. p represents the number of homogenous embeddings.

that contain image preprocessing and some hyperparameters settings. The details of hyperparameter settings are presented as follows. The spatial size $s \times s$ of input image cube for all methods is 27×27. The batch size is adopted as 16. The training epoch is set to 1000. All weight parameters are updated via Adam optimizer with the default settings. In addition, the learning rate is initially set to 0.05 and then halved after 200 epochs. In addition, all experiments are conducted independently for 20 times with randomly sampled sets. In the following, three experiments are reported in detail.

The first experiment is reported on the IP data. The 5% labeled pixels in each land-cover category are randomly selected as training samples, and the remaining ones are as test samples, as presented in Table I. The quantitative classification results are shown in Table VI. Our proposed method achieves the best performance among all methods in terms of the OA and the Kappa coefficient. In particular, it can be observed that H2AN enhances by about 3.7% OA and 4.3% Kappa coefficient compared to baseline HCNN. In contrast, HCNN-SA only increases by 1.5% OA and 2.4% compared to baseline HCNN. Thus, it can be inferred that the good performance significantly benefits from the proposed HAB. Furthermore, the existing sample imbalance leads to unbalanced training for some land cover with few labeled samples. For example, "Grass-p-m" only has one pixel as training sets. The accuracies of HCNN and HCNN-SA are both 0%, while H2AN can obtain 82.6% accuracy for "Grass-p-m." It can be inferred that HAB is superior to SAB in HSI classification particularly for a small number of training sets.

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TABLE VI

CLASSIFICATION ACCURACIES (%) OF THE IP IMAGE OBTAINED BY THE CNN-2D, CNN-3D, CAPSNET, CONVCAPS, SAN, JSSAN, HCNN, HCNN-SA, AND H2AN

Class	CNN-2D	CNN-3D	Conv-Caps	CapsNet	SAN	JSSAN	HCNN	HCNN-SA	H2AN
1	77.8±6.5	43.9±2.3	39.6±8.7	34.6±10.3	72.9±1.9	98.2±1.0	45.7±2.1	80.4±1.1	96.2±0.9
2	82.9±3.4	81.5±2.2	94.4±1.1	89.8±5.6	86.1±3.8	91.4±1.3	91.5±3.4	94.6±0.3	96.0±1.1
3	74.6±1.2	86.5±1.3	87.4±8.5	72.8±5.7	92.4±0.6	95.2±1.9	89.5±1.8	91.4±2.5	95.3±0.8
4	94.0±0.5	93.9±0.7	58.8±0.7	45.5±1.2	89.1±0.1	97.9±0.1	81.9±1.1	88.3±1.1	93.6±0.3
5	81.8±0.4	79.5±1.2	73.4±1.3	79.3±0.3	90.6±0.7	91.8±0.3	93.0±0.5	95.8±0.1	96.9±0.7
6	89.8±0.7	83.8±1.9	92.0±0.4	99.6±0.1	97.1±0.2	96.9±0.2	95.3±0.2	98.5±0.3	96.7±0.3
7	46.2±0.3	26.7±1.3	95.8±1.4	95.5±0.3	47.1±0.4	96.5±0.5	0.0±0.0	0.0±0.0	82.6±0.1
8	88.3±0.8	96.6±1.9	98.2±0.1	99.6±0.0	98.6±0.1	99.8±0.0	99.7±0.1	99.3±0.0	100.0±0.0
9	65.0±0.1	23.1±2.9	52.6±0.2	52.6±0.1	0.0±0.0	70.0±0.8	0.0±0.0	20.4±0.1	51.6±1.1
10	82.0±1.0	87.2±2.3	94.4±1.9	84.6±0.2	91.2±0.1	95.9±0.0	93.5±0.1	95.1±0.4	96.4±0.2
11	87.2±1.7	86.6±1.5	98.9±0.1	94.4±0.1	94.0±0.4	99.2±0.1	96.3±0.1	98.2±0.3	98.5±0.4
12	78.2±2.7	83.4±1.8	97.6±0.1	90.1±0.9	88.8±0.3	90.2±0.0	87.7±0.7	88.8±0.1	96.1±0.5
13	98.6±0.1	94.0±0.7	100.0±0.0	99.5±0.0	98.4±0.1	100.0±0.0	98.3±0.0	98.8±0.0	99.5±0.1
14	95.9±0.0	94.9±0.6	98.7±0.1	98.9±1.5	97.1±0.2	98.5±0.1	97.5±0.5	98.5±0.0	98.8±0.0
15	91.6±1.3	89.4±1.8	67.7±0.4	83.2±2.3	91.8±0.1	97.9±0.8	94.0±0.1	96.0±0.5	98.7±0.6
16	60.0±0.7	73.4±1.7	73.0±0.1	100.0±0.0	87.7±0.4	79.7±0.1	87.2±0.8	80.2±0.1	96.9±0.7
OA	85.7±1.8	86.6±1.3	93.8±2.6	90.0±2.8	93.4±1.5	96.1±0.7	93.5±1.4	95.6±0.8	**97.2±0.5**
Kappa	83.7±2.1	84.6±1.9	92.7±3.5	82.5±3.1	91.4±1.8	95.5±1.2	92.6±1.6	95.0±1.0	**96.9±0.6**

TABLE VII

CLASSIFICATION ACCURACIES (%) OF THE PU IMAGE OBTAINED BY THE CNN-2D, CNN-3D, CAPSNET, CONVCAPS, SAN, JSSAN, HCNN, HCNN-SA, AND H2AN

Class	CNN-2D	CNN-3D	Conv-Caps	CapsNet	SAN	JSSAN	HCNN	HCNN-SA	H2AN
1	89.6±1.1	92.6±2.3	92.5±1.1	97.8±0.1	93.3±0.1	92.6±0.1	93.2±0.5	80.4±1.3	97.0±0.3
2	94.0±0.2	96.5±0.1	98.4±1.4	99.4±0.3	98.4±0.2	99.2±0.1	98.2±0.4	94.6±0.5	99.6±0.1
3	76.6±1.5	83.8±0.7	67.2±0.3	84.0±2.1	94.7±0.4	85.0±0.3	83.9±0.7	91.4±0.7	91.8±0.7
4	84.6±3.6	83.3±0.2	97.0±0.4	97.2±0.4	74.7±0.2	92.6±1.3	83.9±1.1	88.3±0.4	94.0±0.1
5	86.3±1.1	90.0±1.5	100.0±0.0	100.0±0.0	91.2±0.1	100.0±0.0	90.9±0.8	95.8±0.2	95.0±0.7
6	84.1±2.2	93.3±0.9	100.0±0.0	94.8±0.6	98.0±0.3	99.9±0.0	95.6±1.6	98.5±0.1	98.5±0.9
7	82.2±2.9	86.4±1.6	82.4±0.1	0.0±0.0	95.8±0.5	75.9±0.8	87.0±1.5	0.0±0.0	99.1±0.3
8	88.5±0.2	90.2±0.4	92.0±0.1	90.7±0.1	96.7±0.2	96.4±0.3	93.6±1.9	99.3±0.2	94.9±1.3
9	84.03±0.5	66.8±0.8	98.8±0.1	100.0±0.0	63.4±1.6	82.1±0.1	81.7±2.3	20.4±1.7	87.3±1.6
OA	89.3±1.5	92.3±1.1	95.0±1.0	93.6±0.5	94.6±0.3	95.8±0.9	94.1±0.3	95.6±0.5	**97.4±0.6**
Kappa	85.9±1.8	90.0±1.5	92.0±1.3	91.7±0.7	92.9±0.4	94.4±1.1	92.1±0.5	95.0±0.8	**96.5±0.8**

The second experiment is evaluated on the PU dataset. The 1% labeled pixels in each land-cover category are randomly selected as training samples, and the remaining are randomly selected as test samples, as shown in Table I. The quantitative comparisons for different methods are tabulated in Table VII. It can be seen that our proposed H2AN obtains the first place compared to all comparisons in the classification performance. In addition, it can be seen that H2AN not only enhances by 3.3% OA and 4.4% Kappa coefficient but also increases the classification accuracies of all nine land covers compared to HCNN. Furthermore, H2AN outperforms SAN and JSSAN, which are both based on SAB. Thus, it can be inferred that the proposed HAB can extract more discriminative attention features in HSI classification compared to SAB in particular with limited labeled samples as training sets.

The third experiment is conducted on the PC. The 2% of the labeled pixels in each land-cover category are randomly selected as training samples, and the remaining are randomly selected as test samples, which is illustrated in Table I. The classification accuracy per class, OA, and the Kappa coefficient are shown in Table VIII. Similar to the above two datasets, H2AN obtains the first place compared to all compared methods in the classification performance. It can be observed that classification accuracy per class of the proposed H2AN is more than 95.5%, and the classification accuracies of six classes reach 99%. It is noteworthy that H2AN enhances the accuracy of all nine classes of land cover compared to the baseline model, namely, HCNN. In addition, the proposed H2AN model outperforms HCNN-SA in terms of the OA and the Kappa coefficient. There is a reasonable prospect that

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TABLE VIII

CLASSIFICATION ACCURACIES (%) OF THE PAVIA CENTER IMAGE OBTAINED BY THE CNN-2D, CNN-3D, CAPSNET, CONVCAPS, SAN, JSSAN, HCNN, HCNN-SA, AND H2AN

Class	CNN-2D	CNN-3D	Conv-Caps	CapsNet	SAN	JSSAN	HCNN	HCNN-SA	H2AN
1	99.9±0.0	99.8±0.1	100.0±0.0	99.9±0.0	99.9±0.1	99.8±0.0	99.9±0.1	99.9±0.1	100.0±0.0
2	62.7±1.3	87.2±0.5	96.9±0.4	99.7±0.1	86.6±0.7	94.4±0.3	94.5±0.7	96.6±0.4	98.1±0.3
3	53.5±7.4	55.4±2.3	83.3±0.2	39.1±0.1	77.6±0.4	86.3±0.7	83.3±0.4	91.1±0.7	95.6±0.0
4	57.0±6.3	42.0±3.6	69.3±0.4	72.0±0.3	96.9±0.7	98.6±0.1	97.9±0.3	98.8±0.3	99.5±0.1
5	73.7±0.3	85.9±1.3	96.2±0.1	96.0±0.5	98.1±0.4	97.1±0.8	97.3±1.3	98.5±1.3	99.9±0.0
6	78.7±0.7	81.4±0.1	99.5±0.3	99.8±0.1	95.5±0.3	98.2±0.1	97.3±0.7	98.4±0.2	99.0±0.3
7	89.1±0.1	90.4±1.3	84.4±0.4	84.0±0.9	99.1±0.7	97.9±0.3	98.2±0.5	99.6±0.4	99.2±0.1
8	97.1±0.6	98.8±0.4	99.9±0.0	99.3±0.7	99.0±0.2	96.1±1.3	99.3±1.3	99.5±0.7	99.9±0.0
9	67.2±0.1	81.9±0.1	97.2±1.3	100.0±0.0	81.8±0.7	90.5±0.3	88.9±1.3	93.4±0.6	97.2±0.4
OA	91.7±1.6	94.5±0.8	97.8±0.1	97.0±0.3	97.7±0.3	98.6±0.4	98.5±0.5	99.2±0.4	**99.7±0.1**
Kappa	88.2±2.3	92.2±0.4	97.0±0.3	96.6±0.5	96.7±0.4	97.7±0.6	97.9±0.9	98.8±0.7	**99.4±0.2**

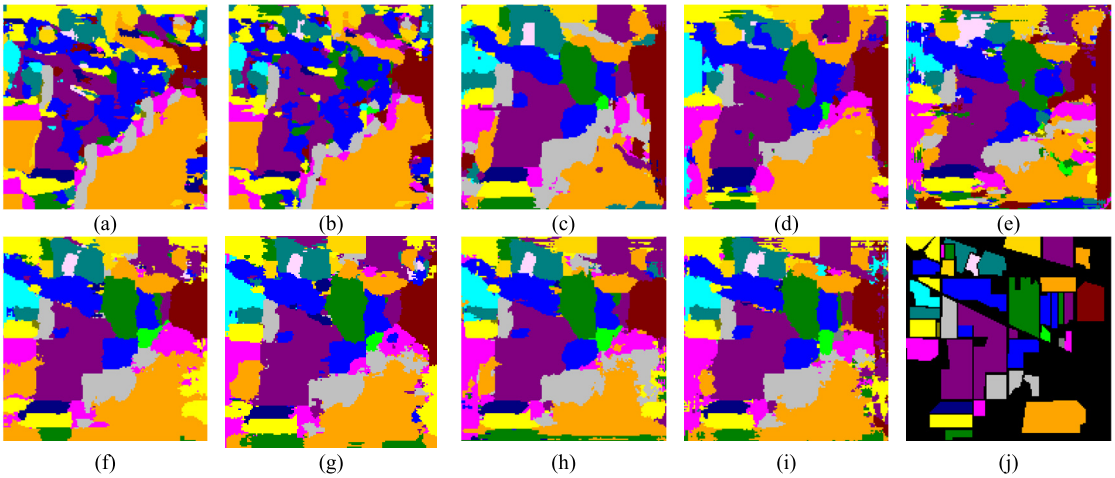


Fig. 9. Classification maps obtained by using different classification methods for the Indian Pines dataset. (a) Conv-2D. (b) CNN-3D. (c) Conv-Caps. (d) CapsNet. (e) SAN. (f) JSSAN. (g) HCNN. (h) HCNN-SA. (i) H2AN. (j) Ground truth.

HAB can capture beneficial features for classification more effectively than the SAB especially if the number of training sets is few.

For qualitative evaluation, the classification maps obtained by corresponding methods on three datasets are visually compared from Figs. 9 to 11. Compared to the other methods, the visualization results of H2AN are nearest to the ground truth on three datasets. Especially, on the Indian dataset, it can be observed that land covers are arranged closely, which gives difficulty in the classification of class boundaries. The result of H2AN is smoother than those of all comparative methods on the IP dataset. It can be inferred that the good performance on the classification of mixed pixels around the class boundaries benefits from the efficient ability to capture discriminative attention features given by HAB. On the PU dataset and the PC, the visualization results of comparative methods are not smooth on account of only 1% labeled samples and 2% labeled samples as training sets. In particular, those attention CNN-based methods using original SAB (SAN, JSSAN, and HCNN-SA) achieve poor performance on the edges of land covers. However, it can be clearly seen that our proposed H2AN effectively suppresses the effects of interfering pixels and smoothens most classification results of class boundary pixels on the classification maps.

To sum up, our H2AN obtains better performance than all the compared state of the arts in the following three aspects: 1) good classification accuracy of the whole image on overall test HSI datasets in OA index; 2) robust class accuracy of different land covers in the Kappa coefficient, which can overcome the class-imbalance problem; and 3) smooth effect in visual classification map on either pure or mixed pixels. Based on the above quantitative and qualitative analysis, our H2AN can obtain solid classification results on land covers no matter what their category is or how their class number and spatial distribution are.

D. Visualization of Feature Maps

From the above experimental contrast analysis, the performance of H2AN is satisfactory, which is mostly attributed to the introduction of effective HAB's feature extraction mode. To furthermore demonstrate that the proposed H2AN can

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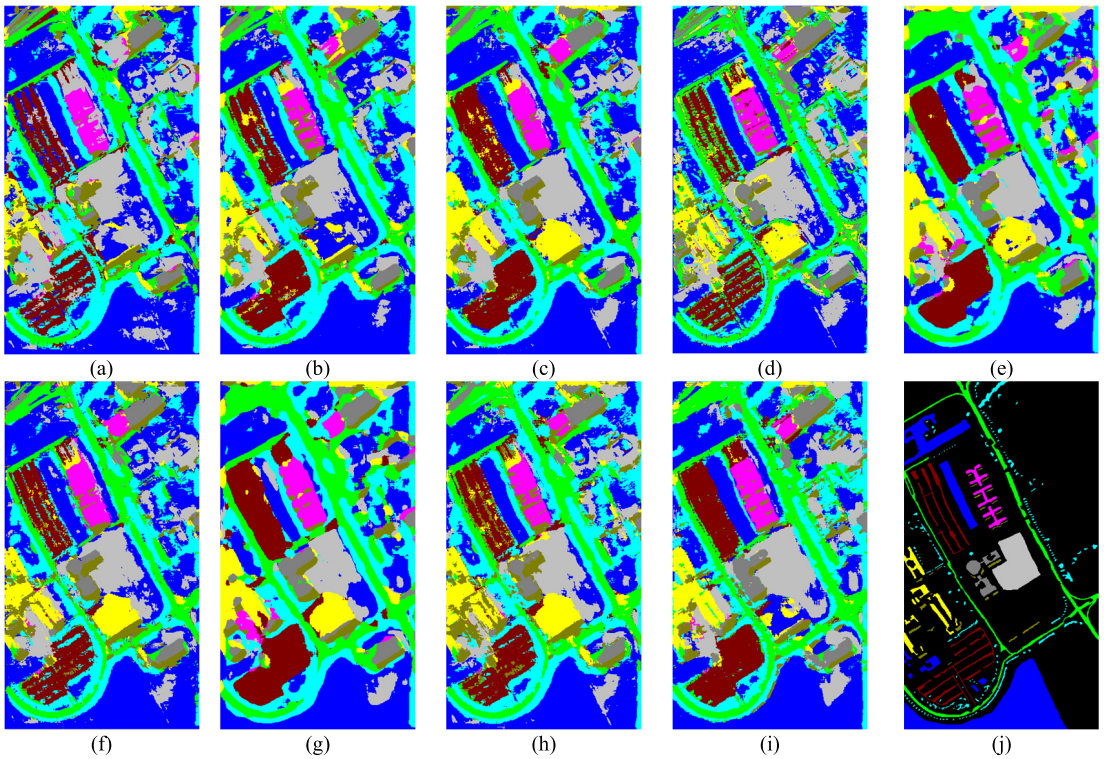


Fig. 10. Classification maps obtained by using different classification methods for PU. (a) Conv-2D. (b) CNN-3D. (c) Conv-Caps. (d) CapsNet. (e) SAN. (f) JSSAN. (g) HCNN. (h) HCNN-SA. (i) H2AN. (j) Ground truth.

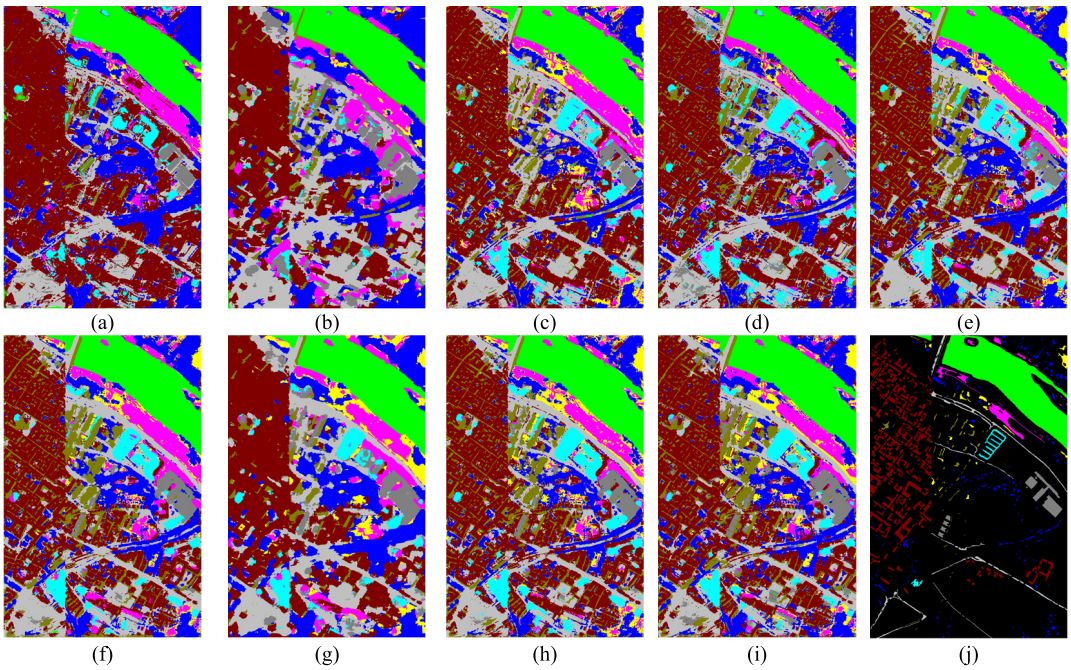


Fig. 11. Classification maps obtained by using different classification methods for the PC. (a) Conv-2D. (b) CNN-3D. (c) Conv-Caps. (d) CapsNet. (e) SAN. (f) JSSAN. (g) HCNN. (h) HCNN-SA. (i) H2AN. (j) Ground truth.

focus on the discriminative feature extraction of the attention areas effectively, the CAM [38] is implemented to obtain the qualitative analysis. CAM is a recently proposed visualization method that obtains class activation maps by a linear combination of activation maps and weights obtained by its forward passing score on the target class. The class activation map can clearly present the target regions with respect to a specific class. The attention regions are considered important parts for predicting the target class of land cover. Thus, we attempt to find whether the proposed H2AN can focus on attention areas

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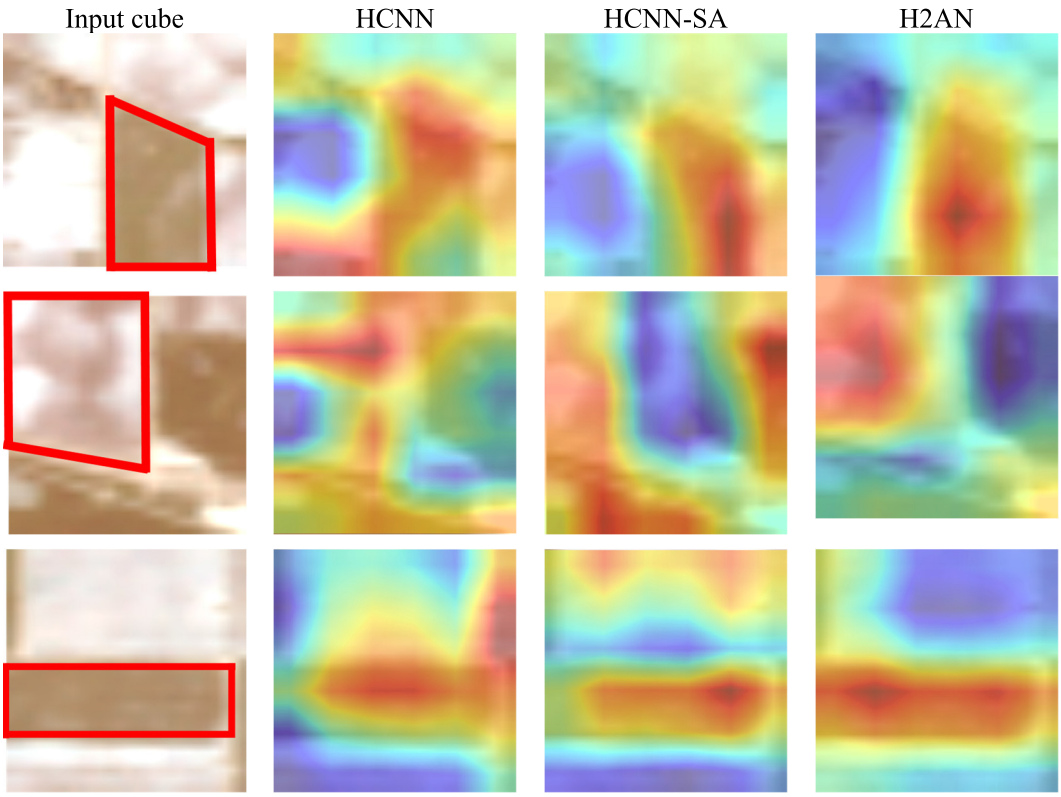


Fig. 12. CAM visualization results for the IP dataset. We compare the visualization results of H2AN with baseline (HCNN) and HCNN-SA. The attention areas of the input cubes are marked by the red lines, as shown in the input cube. The highlighted red regions in the heat maps denote the discriminative features for classification extracted by the corresponding methods.

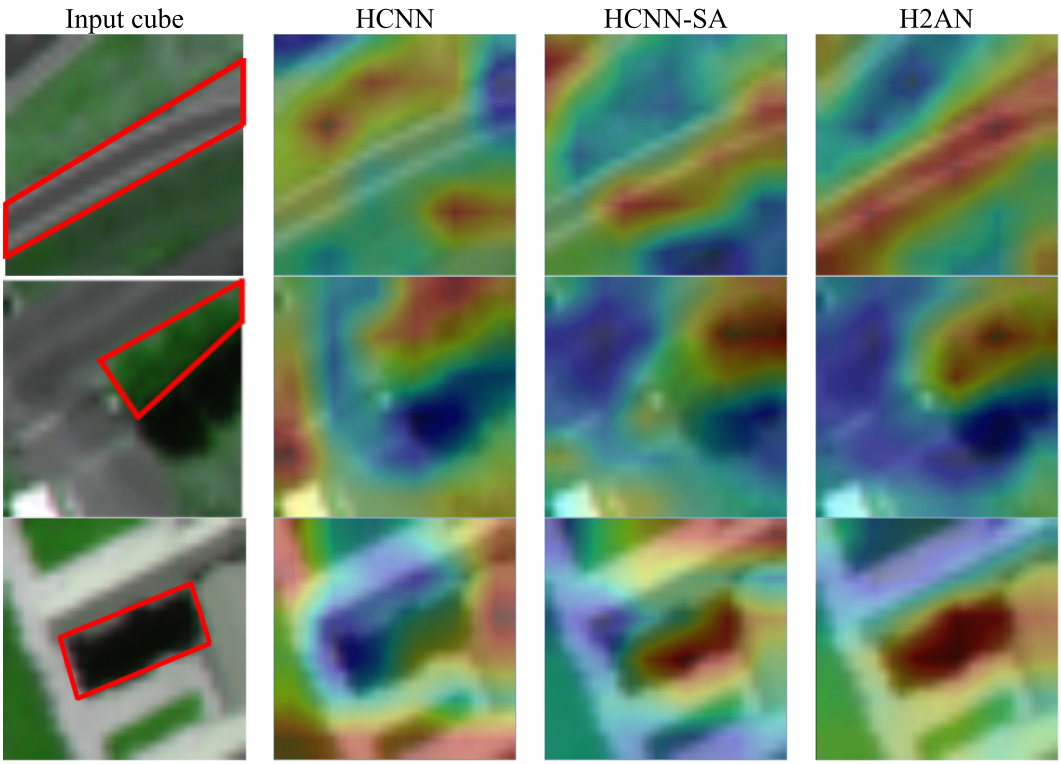


Fig. 13. CAM visualization results for the PU dataset. We compare the visualization results of H2AN with baseline (HCNN) and HCNN-SA. The attention areas of the input cubes are marked by the red lines, as shown in the input cube. The highlighted red regions in the heat maps denote the discriminative features for classification extracted by the corresponding methods.

and alleviate the effects of interfering pixels. We compare three mixed image cubes' visualization results of H2AN with baseline (HCNN) and HCNN-SA on each dataset, which is shown in Figs. 12–14. It can be clear that the CAM masks

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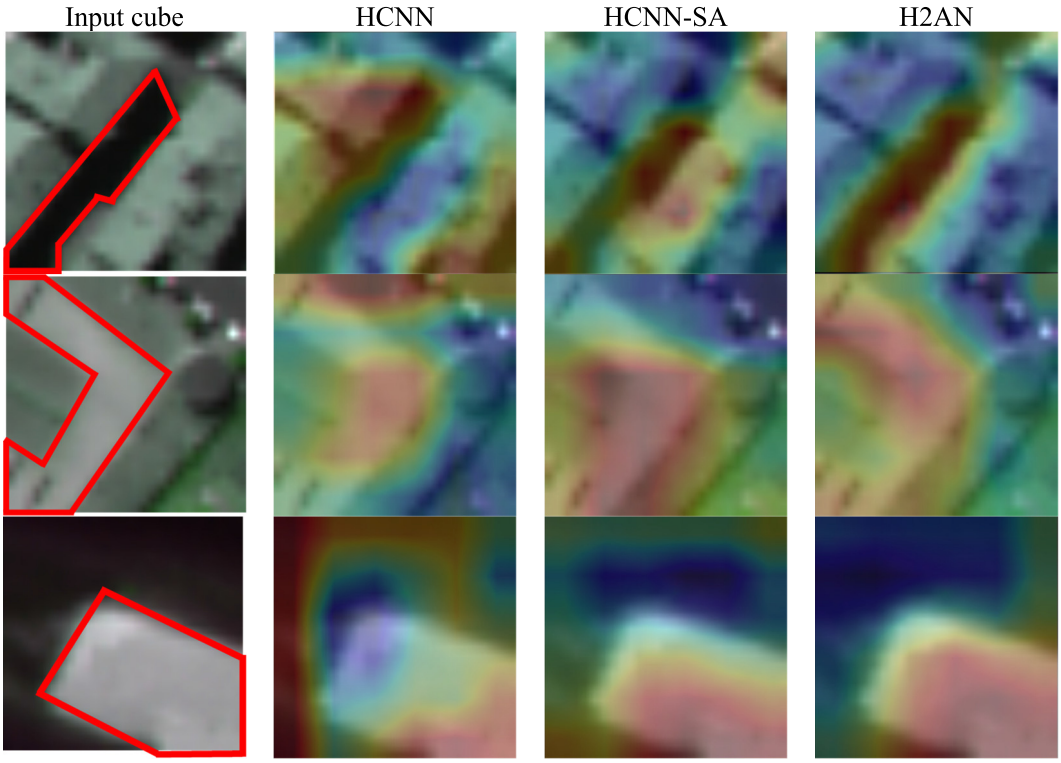


Fig. 14. CAM visualization results for the PC. We compare the visualization results of H2AN with baseline (HCNN) and HCNN-SA. The attention areas of the input cubes are marked by the red lines, as shown in the input cube. The highlighted red regions in the heat maps denote the discriminative features for classification extracted by the corresponding methods.

TABLE IX
ELAPSED TIME OF DIFFERENT METHODS ON THE IP DATASET

Datasets	Method	Training time (s)	Test time (s)
Indian Pines	CNN-2D	285.8±15.3	11.2±6.4
	CNN-3D	343.6±24.6	17.5±0.3
	Conv-Caps	589.1±30.7	50.9±2.9
	CapsNet	905.8±27.9	80.8±5.6
	SAN	580.9±15.1	25.9±0.5
	JSSAN	543.2±21.6	23.7±0.4
	HCNN	273.5±18.2	12.8±0.2
	HCNN-SA	455.6±23.7	18.3±0.6
	H2AN	392.4±15.5	14.9±0.1

TABLE X
ELAPSED TIME OF DIFFERENT METHODS ON THE PU DATASET

Datasets	Method	Training time (s)	Test time (s)
Pavia University	CNN-2D	66.3±7.6	26.1±0.9
	CNN-3D	96.4±6.5	35.3±4.3
	Conv-Caps	213.2±7.3	361.4±5.9
	CapsNet	1257.1±9.3	450.6±7.4
	SAN	722.3±7.1	70.6±5.7
	JSSAN	665.1±12.4	65.3±1.6
	HCNN	77.8±1.2	28.2±0.9
	HCNN-SA	217.4±1.5	48.1±0.6
	H2AN	205.6±2.6	42.3±0.3

TABLE XI
ELAPSED TIME OF DIFFERENT METHODS ON THE PC

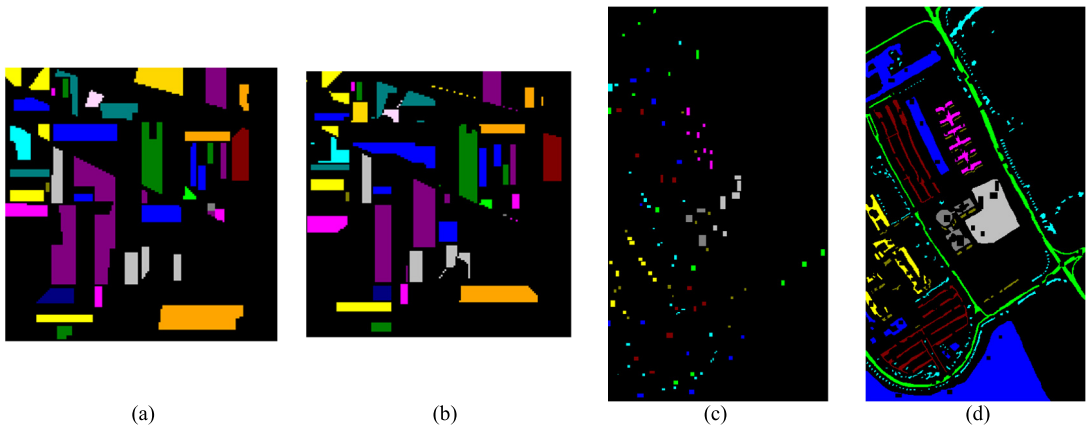
Datasets	Method	Training time (s)	Test time (s)
Pavia Center	CNN-2D	131.1±6.2	69.0±3.6
	CNN-3D	643.7±7.4	145.2±2.4
	Conv-Caps	1209.5±9.3	679.5±5.6
	CapsNet	1603.4±11.1	900.8±6.4
	SAN	1088.9±9.8	294.3±2.7
	JSSAN	876.1±0.9	223.5±0.1
	HCNN	283.9±26.7	89.9±0.5
	HCNN-SA	662.6±28.8	163.2±1.1
	H2AN	473.7±12.5	125.3±0.4

of H2AN cover the target object regions better than other methods on three datasets. Because the background pixels are complex and the number of labeled training samples is few, HCNN and HCNN-SA are both not able to effectively cover the target region, which is shown in their scattered heat map results. As a contrast, the H2AN learns well to exploit important information in target object regions and aggregate discriminative features from them. It can be inferred the hierarchical HABs enhance the proposed network the ability to locate the target regions and suppress the effects of interfering pixels. Thus, the proposed H2AN obtains better classification results than the comparisons.

E. Running Time Comparison

To further demonstrate the lower consumption complexity of the proposed method, we have conducted the corresponding experiments on three datasets. Tables IX–XI show the execution time in the training and test process for all the algorithms on three datasets. From Tables IX–XI, we can obtain that our method requires limited calculated cost in three aspects.

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(a) (b) (c) (d)

Fig. 15. Spatially disjoint sampling distributions on IP and PU datasets. (a) Disjoint IP train. (b) Disjoint IP test. (c) Disjoint PU train. (d) Disjoint PU test.

TABLE XII
OA RESULTS OBTAINED BY DIFFERENT METHODS ON TWO DISJOINT DATASETS

Method	CNN-1D[39]	CNN-2D	CNN-3D	Conv-Caps	CapsNet	SAN	JSSAN	HCNN	HCNN-SA	H2AN
IP(%)	84.94	59.60	78.56	80.53	77.67	82.35	84.23	68.81	71.56	**87.00**
PU(%)	87.59	75.34	86.37	85.64	83.29	85.12	86.69	83.56	84.01	**88.65**

1) Without the time-consuming dynamic routing algorithm for the calculation between capsules, the CNN-based deep learning methods, including our method, need less time to train and test the networks than capsule-based methods (e.g., Conv-caps and CapsNet).
2) Because of the additional attention processing, the SAB-based methods require more time than the baseline HCNN method. Among them, HCNN-SA only considers the spatial SAB so that it spends the lowest training and test time.
3) Due to our enhancement of SAB in the attention learning process, our H2AN has lower time complexity in training and testing stages. Especially, it can reduce 3.4-, 5.8-, and 37.9-s test times for, respectively, IP, PU, and PC datasets compared with HCNN-SA.

F. Investigation on Training and Testing With Spatially Disjoint Samples

Paoletti *et al.* [39] pointed that new trends suggest that the high correlation between neighboring pixels can affect the performance of the network. In this section, our proposed method and the abovementioned comparisons are trained and tested on two disjoint datasets, namely, spatially disjoint samples of IP and PU datasets [40]. The sampling distributions of the above two datasets are shown in Fig. 15. It can be obviously seen that this sampling method avoids the possible overlapping effects between the test and train sets. Table XII shows the OA results obtained by different methods. Particularly, in [39], the best OA result for the IP dataset is 85.08% with traditional SVM and 84.94% with the CNN1D deep learning-based method. For the PU dataset, the best OA result is 81.96% with traditional multi-layer perceptron (MLP) and 87.59% with the CNN1D deep learning-based method. From Table XII, we can observe the following.

1) The disjoint setting of the training and testing sample makes the deep learning-based methods that depend on spatial information inferior, but our method obtains comparative performance because of homogeneous attention. The proposed method achieves the best classification performance on both of these two disjoint datasets compared with the best methods in [39] and other our previous selected compared approaches including traditional CNN-based methods (CNN-2D and CNN-3D), Capsule-based methods (Conv-caps and CapsNet), and representative self-attention-based methods (SAN and JSSAN).
2) Our H2AN can obtain better performance than the baseline HCNN and the related HCNN-SA due to the combination of HAB and HCNN. Compared with the baseline HCNN, the performance of H2AN enhances by 18.19% OA on the IP dataset and 5.09% OA on the PU dataset, while HCNN-SA only increases by 2.75% OA on the IP dataset and 0.45% OA on the PU dataset.

IV. CONCLUSION

In this article, a novel H2AN is proposed for HSI classification. Different from the traditional SAN that lacks position supervision of target regions and computes similarity matrixes inefficiently, our H2AN is committed to concentrating on the highly efficient feature extraction of object regions in mixed pixels. Specifically, HAB is first designed to depict the feature correlation via a homogeneous mask. The predefined number of homogeneity embeddings is selected as the similarity comparisons using the homogeneous mask, which can reduce the heavy computation cost of the following attention process. Finally, a simple CNN is embedded with HAB hierarchically to transfer the supervision from shallow to deep layer, which

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can, furthermore, enhance the ability to extract the features of target regions. The typical experimental results on three widely used HSI datasets demonstrate that the proposed H2AN is able to yield better performance compared with the state-of-the-art methods. In addition, the exciting CAM visualization results show that H2AN can, indeed, concentrate on the target regions and ignore the effects of interfering pixels intuitively, which can be a rational explanation for its satisfactory performance.

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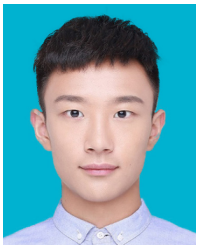
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